EE3621: Digital Signal Processing

Lecture 22: Sampling Theorem, Aliasing & A/D and D/A Conversion
(III B.Tech EEE)

Lecture Notes (40-Minute Plan)

1 Lecture Plan & Objectives (40 Minutes Breakdown)

- **00:00 – 05:00 (5 mins):** Introduction to Continuous-Time Sampling & Mathematical modeling.
- **05:00 – 12:00 (7 mins):** Nyquist-Shannon Sampling Theorem: Formal statement, proof, and conditions.
- **12:00 – 17:00 (5 mins):** Aliasing phenomenon, folding frequency, and numerical example.
- **17:00 – 22:00 (5 mins):** Practical ADC Pipeline: Anti-aliasing, Sample-and-Hold, Quantization.
- **22:00 – 27:00 (5 mins):** Reconstruction (DAC): Zero-Order Hold (ZOH) droop and interpolation filters.
- **27:00 – 31:00 (4 mins):** Oversampling ADC: SNR improvement and quantization noise spreading.
- **31:00 – 35:00 (4 mins):** Sigma-Delta ADC: First-order and second-order noise shaping.
- **35:00 – 38:00 (3 mins):** Sample Rate Conversion: Rational L/M factor and polyphase implementation.
- **38:00 – 40:00 (2 mins):** Checkpoints, Q&A, and conclusion.

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2 Continuous-Time Sampling

2.1 Physical Intuition

Sampling is the process of taking a continuous-time signal and measuring its value at discrete time intervals. Think of it like taking a stroboscopic video of a moving fan. If you take pictures fast enough, you capture the true motion. If too slow, the fan might appear to spin backward!

2.2 Mathematical Model: Ideal Sampling

We model the ideal sampling of a continuous-time signal $x(t)$ by multiplying it with an impulse train (or Dirac comb) $p(t)$.

1. Let T be the sampling period.
2. The sampling frequency is $f_s = 1/T$.

3. The impulse train is defined as:

$$p(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT) \tag{1}$$

4. The sampled signal $x_s(t)$ is the product:

$$x_s(t) = x(t) \cdot p(t) = \sum_{n=-\infty}^{\infty} x(nT)\delta(t - nT) \tag{2}$$

2.3 Frequency Domain Analysis (Derivation)

To understand what sampling does to the frequency content, we take the Fourier transform of $x_s(t)$.

1. The Fourier transform of a product in time is a convolution in frequency:

$$X_s(f) = X(f) * P(f) \tag{3}$$

2. The Fourier transform of the impulse train $p(t)$ is another impulse train in the frequency domain:

$$P(f) = \frac{1}{T} \sum_{k=-\infty}^{\infty} \delta\left(f - \frac{k}{T}\right) = f_s \sum_{k=-\infty}^{\infty} \delta(f - kf_s) \tag{4}$$

3. Substituting $P(f)$ into the convolution equation:

$$X_s(f) = X(f) * \left(f_s \sum_{k=-\infty}^{\infty} \delta(f - kf_s)\right) \tag{5}$$

4. Using the distributive property of convolution and the sifting property of the Dirac delta function ($X(f) * \delta(f - f_0) = X(f - f_0)$):

$$X_s(f) = f_s \sum_{k=-\infty}^{\infty} X(f - kf_s) \tag{6}$$

KEY RESULT: The spectrum of the sampled signal, $X_s(f)$, consists of periodically repeating copies (images) of the original baseband spectrum $X(f)$, scaled by f_s and shifted by integer multiples of the sampling frequency f_s.

3 Nyquist-Shannon Sampling Theorem

3.1 Formal Statement

If a continuous-time signal $x(t)$ is bandlimited to f_{max} (meaning $X(f) = 0$ for $|f| > f_{max}$), then $x(t)$ can be perfectly reconstructed from its discrete samples $x(nT)$ if and only if the sampling frequency f_s satisfies:

$$f_s \geq 2f_{max} \tag{7}$$

The minimum required sampling rate, $2f_{max}$, is called the **Nyquist rate**.

3.2 Proof of Reconstruction

1. If $f_s \geq 2f_{max}$, the spectral copies in $X_s(f)$ do not overlap. The baseband copy is perfectly preserved in the interval $[-f_s/2, f_s/2]$.
2. To recover $x(t)$, we apply an ideal lowpass filter (brick-wall filter) $H(f)$ with a gain of T and a cutoff frequency f_c such that $f_{max} \leq f_c \leq f_s - f_{max}$. Typically, we choose $f_c = f_s/2$.
3. The frequency response of this ideal filter is:

$$H(f) = T \cdot \text{rect}\left(\frac{f}{f_s}\right) \tag{8}$$

4. The recovered spectrum is $X_r(f) = X_s(f)H(f) = X(f)$.
5. In the time domain, multiplication in frequency becomes convolution in time. The inverse Fourier transform of $H(f)$ is the sinc function:

$$h(t) = \text{sinc}(f_s t) = \frac{\sin(\pi f_s t)}{\pi f_s t} \tag{9}$$

6. The reconstructed signal is:

$$x(t) = x_s(t) * h(t) = \left(\sum_{n=-\infty}^{\infty} x(nT)\delta(t - nT)\right) * \text{sinc}(f_s t) \tag{10}$$

7. Evaluating the convolution yields the ideal sinc interpolation formula:

$$x(t) = \sum_{n=-\infty}^{\infty} x[n]\text{sinc}(f_s(t - nT)) = \sum_{n=-\infty}^{\infty} x[n]\text{sinc}(f_s t - n) \tag{11}$$

4 Aliasing

4.1 When the Nyquist Criterion is Violated

If $f_s < 2f_{max}$, the shifted spectral copies in $X_s(f)$ will overlap. This overlapping of high-frequency components into the lower-frequency bands is called **aliasing**. Once aliasing occurs, the original signal is irreversibly distorted, and perfect reconstruction is impossible.

4.2 Aliased Frequency Calculation

When a frequency component f is sampled at f_s, it can appear at multiple alias frequencies. The apparent (aliased) frequency in the baseband $[-f_s/2, f_s/2]$ is given by:

$$f_a = |f - kf_s| \tag{12}$$

where k is the integer that brings f_a into the range $[0, f_s/2]$.

4.3 Numerical Example:

Problem: A 3 kHz tone is sampled at 4 kHz. What frequency will be observed in the reconstructed signal?
Calculation:

1. Original frequency: $f = 3$ kHz.

2. Sampling frequency: $f_s = 4$ kHz.

3. Check Nyquist: $2f = 6$ kHz > 4 kHz. Aliasing will occur!

4. Calculate alias for $k = 1$:

$$f_a = |3 \text{ kHz} - 1 \times 4 \text{ kHz}| = |-1| \text{ kHz} = 1 \text{ kHz} \tag{13}$$

Conclusion: The 3 kHz tone masquerades as a 1 kHz tone.

5 Practical ADC Pipeline

A real-world Analog-to-Digital Converter (ADC) requires several stages:

1. **Anti-Aliasing Filter (AAF):** An analog Low-Pass Filter (LPF) placed before sampling to strictly enforce the bandlimit $f_{max} < f_s/2$. Because practical analog filters cannot have a perfect "brick-wall" roll-off, we must sample at a rate somewhat higher than $2f_{max}$ to provide a transition band.

2. **Sample-and-Hold (S/H):** Captures the continuous voltage and holds it steady while the quantizer resolves the bits.

3. **Quantizer:** Maps the continuous voltage level to the nearest discrete level.

4. **Encoder:** Assigns a digital binary code to the quantized level.

6 Reconstruction (DAC)

6.1 Zero-Order Hold (ZOH)

Practical DACs hold the sample value constant for the entire sampling period T.

1. The impulse response of a ZOH is a rectangular pulse:

$$h_{ZOH}(t) = 1 \quad \text{for } 0 \leq t < T \tag{14}$$

2. The frequency response is the Fourier transform of this pulse:

$$H_{ZOH}(f) = T\text{sinc}(fT)e^{-j\pi fT} \tag{15}$$

3. **ZOH Distortion (Droop):** The magnitude $|H_{ZOH}(f)|$ rolls off, attenuating higher frequencies in the baseband.

6.2 Interpolation Filter

To compensate for the ZOH droop and to remove the spectral images at multiples of f_s, a digital or analog **interpolation filter** is used. These filters are often FIR filters designed using the window method.

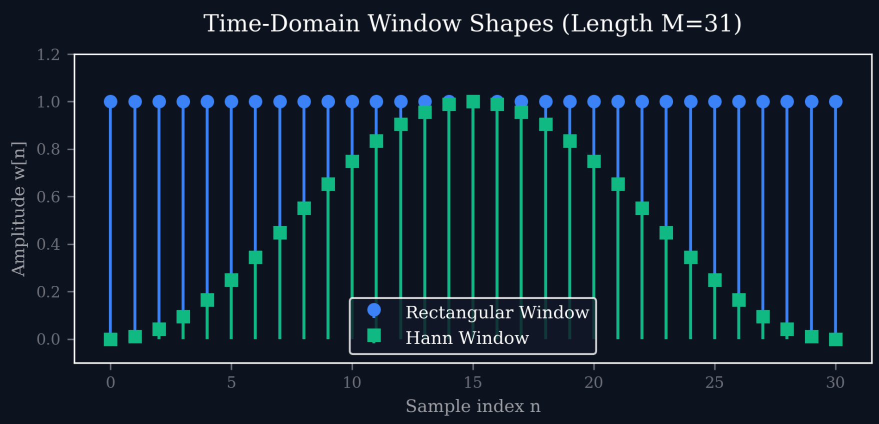

Figure 1: Time-domain shapes of windows used in interpolation filter design.

A poor window choice (like rectangular) results in the Gibbs phenomenon, causing passband ripples and poor image rejection, as shown here:

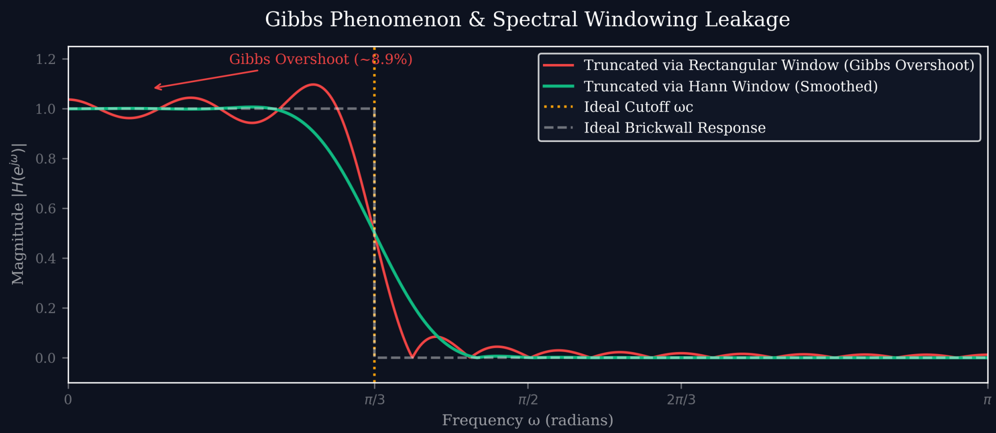

Figure 2: Gibbs Phenomenon: Rectangular vs. Hann window responses.

7 Oversampling ADC

Oversampling means sampling at kf_s, where f_s is the Nyquist rate and $k \gg 1$.

7.1 Benefits of Oversampling

1. **Simplified Analog Anti-Aliasing Filter:** The transition band is massively widened.
2. **Quantization Noise Reduction:** Noise power is spread evenly over $[0, kf_s/2]$.
3. **Digital Filtering:** A digital lowpass filter removes noise outside the baseband $[0, f_s/2]$.
4. **SQNR Improvement:** The Signal-to-Quantization-Noise Ratio (SQNR) improves by:

$$\Delta\text{SQNR} = 10\log_{10}(k)\ \text{dB} \tag{16}$$

8 Sigma-Delta ($\Sigma\Delta$) ADC

Sigma-Delta ADCs combine massive oversampling with **noise shaping**.

8.1 First-Order Noise Shaping

By placing the quantizer inside a feedback loop with an integrator, the quantization noise is high-pass filtered. The noise transfer function (NTF) is $1 - z^{-1}$. First-order shaping provides a 9 dB SQNR improvement per octave.

8.2 Second-Order Noise Shaping

Using two integrators gives a second-order noise shaping:

$$NTF(z) = (1 - z^{-1})^2 \tag{17}$$

This provides a 15 dB SQNR improvement per octave (2.5 bits), allowing a simple 1-bit quantizer to achieve 24-bit audio resolution after decimation!

9 Sample Rate Conversion

9.1 Rational L/M Factor Conversion

If the ratio of the new rate to the old rate is a rational number L/M:

1. **Interpolation:** Insert $L - 1$ zeros between each sample.
2. **Low-Pass Filter:** Apply a digital low-pass filter at the intermediate high rate to remove imaging.
3. **Decimation:** Keep only every M-th sample.

9.2 Polyphase Filter Implementation

The filter is decomposed into L sub-filters (polyphase decomposition). This allows the filtering to happen at the lower sample rates.

10 Key Formulas Table

Concept	**Formula**
Sampled Spectrum	$X_s(f) = f_s \sum_k X(f - kf_s)$
Nyquist Criterion	$f_s \geq 2f_{max}$
Ideal Reconstruction	$x(t) = \sum_n x[n]\text{sinc}(f_s t - n)$
Aliased Frequency	$f_a = \lvert f - kf_s \rvert$
ZOH Frequency Response	$H_{ZOH}(f) = T\text{sinc}(fT)e^{-j\pi fT}$
Oversampling SQNR Gain	$10\log_{10}(k)$ dB

11 Checkpoints & Quick Review Questions

1. **Q1:** An analog signal with frequency components up to 15 kHz is sampled at 20 kHz. Is the Nyquist criterion met? If a 12 kHz component exists in the signal, at what frequency will it appear in the sampled digital signal?
   - *Answer:* The Nyquist criterion requires $f_s \geq 2f_{max}$. Here, $2f_{max} = 30$ kHz, but $f_s = 20$ kHz. Thus, the **Nyquist criterion is NOT met**. To find the aliased frequency of the 12 kHz tone, we use $f_a = |f - kf_s|$. For $k = 1$: $f_a = |12 \text{ kHz} - 1(20 \text{ kHz})| = |-8 \text{ kHz}| = 8$ kHz. The 12 kHz tone will appear as an **8 kHz alias**.

2. **Q2:** Explain physically why a Zero-Order Hold (ZOH) circuit introduces distortion (droop) in the reconstructed signal's passband, and how a digital interpolation filter can correct this.

    - *Answer:* Physically, holding a sample constant for duration T creates a stair-step waveform. The flat part essentially acts as a low-pass averager in the time domain, which attenuates the higher frequencies of the original baseband signal. Mathematically, this is the $\text{sinc}(fT)$ frequency response. An interpolation filter can correct this by applying an "inverse-sinc" equalization in its passband to perfectly cancel out the ZOH attenuation before sharply cutting off to remove images.

3. **Q3:** Why does a Sigma-Delta ADC use a 1-bit quantizer instead of a multi-bit quantizer, given that 1-bit quantization introduces massive quantization noise?

    - *Answer:* A 1-bit quantizer is perfectly linear, eliminating the differential non-linearity (DNL) errors inherent in multi-bit resistor ladders. While 1-bit quantization adds huge noise, the feedback loop in the $\Sigma\Delta$ architecture performs **noise shaping**, pushing this noise to very high frequencies outside the signal band. Massive **oversampling** combined with a digital low-pass decimation filter completely removes this high-frequency noise, leaving an incredibly high Signal-to-Noise Ratio.